New Service Teams as Information-Processing Systems

Reducing Innovative Uncertainty

Annouk Lievens
Maastricht University

Rudy K. Moenaert
University of Ghent

This article theoretically and empirically examines the antecedents and consequences of project communication during the new financial service innovation process. The authors analyze project communication comprising both intraproject communication and extraproject communication (i.e., boundary spanners) and adopt the view that project teams within banks are primarily information-processing systems directed toward reducing innovative uncertainty. The research findings indicate that the level of complexity contributes to intraproject communication, whereas centralized project environments appear to be a barrier for communication within the project team. Curvilinear relationships (inverted U) are substantiated between project climate and intraproject communication and between formalization and boundary-spanning communication. The authors' findings provide support for the fact that effective project communication is contingent upon the level of cross-functional cooperation and that the relationship with project success is an indirect one, mediated by the level of innovative uncertainty reduction.

The study of innovation in the financial service industry is a relatively new research topic in marketing and adds to existing contributions in the service innovation literature. Financial industries have changed rapidly. Not only have new market opportunities emerged (i.e., banks now also engage in insurance and real estate services), but especially, new technology has proliferated. The latter has strongly increased the rate of innovation. Telecommunication systems facilitate paper flows, make direct interactions with customers possible, and foremost, make the engineering of new products (such as loans, securities, foreign exchange, etc.) possible within days, sometimes even hours (Drew 1994). Innovation in financial services comprises a wide range of forms in which new products and new processes may be distinguished. Customers buy *products* from the organization, whereas *process* refers to an organized system for delivering one or more products (Drew 1994; Macdonald 1998). This categorization is often difficult. Home banking and office banking are new products, but they also represent changes in the delivery processes and systems of the banks. Moreover, some new processes may seem to have little direct impact on the customer (e.g.,

A preliminary version of this article was presented as a work-in-progress paper at the annual EGOS conference, Maastricht, in June 1998. We greatly appreciate the comments of Henry Robben, Piet Vanden Abeele, Filip Caeldries, Marc Jegers, and Roland Pepermans on earlier drafts of this article. Also, the comments from the journal editor and reviewers are gratefully acknowledged.

Journal of Service Research, Volume 3, No. 1, August 2000 46-65

additional and new administrative procedures or software enhancement). However, even in these cases, indirect benefits are sought in terms of faster, less expensive, or more accurate customer service (Lievens, de Ruyter, and Lemmink 1999).

Recent innovation studies have provided insight into the factors responsible for new financial service success and failure (e.g., de Brentani 1989, 1991, 1993b, 1995b; de Brentani and Cooper 1991, 1992; Easingwood 1986; Easingwood and Percival 1990; Easingwood and Storey 1991, 1993, 1994). Despite the contribution of these studies to our understanding of innovation in the financial services industry, the role of communication has largely been ignored in this research area. Only a few studies have indicated that the effectiveness of internal communication (de Brentani 1989, 1993b, 1995b) and external communication (Cooper et al. 1994; Easingwood and Storey 199b) is indeed a critical condition for new service success. Communication by project team members is crucial because it provides adequate market information on competitors and customers. Also, technical information is often channeled into the project team through cooperation with affiliated companies. Moreover, communication does not only help the bank to channel external information into the project team. Communication and crbss-functional cooperation is necessary to effectively integrate new banking services into the existing product range and information technology infrastructure (Lievens, Moenaert, and S'Jegers 1999).

Thus, we know that effective communication is essential to new financial service success, but we still lack an in-depth understanding of the antecedents and the effects of communication flows during the innovation process of these services. This article will address the following questions: How effective are these communication flows in reducing the level of innovative uncertainty? What are the conditions in organizational design that enhance and impede communication flows during the project life cycle? How does the level of cross-functional cooperation moderate the relationship between communication and innovative uncertainty? Finally, to what extent does effective communication (i.e., the reduction of innovative uncertainty) enhance project success (i.e., the performance of the new financial service)? First we present our theoretical model, and then we discuss our research design, which has been developed to test the developed hypotheses. We conclude this article by discussing the research results and by evaluating the managerial implications of this study.

AN INFORMATION-PROCESSING APPROACH

During service production and delivery, there almost always is some level of customer involvement, which gives rise to strong functional interdependencies within the financial service organization. Simultaneous production and consumption cannot be separated due to the presence of the customer as part of the system; therefore, service organizations cannot separate production and marketing activities. This creates a potential problem because, for most financial institutions, the individuals responsible for producing the service (i.e., the operations function) are different from those who deal with the customer (i.e., the marketing function). In other words, there is a functional separation between the so-called back office (operations) and front office (customer contact). Consequently, the interface or interaction between these functions becomes critical for a successful service offering (Chase 1981; Chase and Tansik 1983; Lovelock et al. 1988; Mahajan et al. 1994).

This functional interdependence between operations and marketing has an important impact on the service innovation process. Because of the required work and information flows between these highly interdependent organizational functions, communication between departments becomes crucial during service development. These organizational entities are in turn interdependent on some larger environments (Mahajan et al. 1994; Ruekert and Walker 1987; Thompson 1967). Thus, in line with an open-system view (see Allen 1985; Brown and Utterback 1985; Daft and Weick 1984; Ebadi and Utterback 1984; Fidler and Johnson 1984; Moenaert and Souder 1990b; Smith et al. 1994; Thompson 1967; Tushman 1979a, 1979b, 1979c; Weick 1969), uncertainty is introduced to these work-related information processes. Consequently, the innovation process can be viewed as a process of uncertainty reduction as the financial service project team deals with work-related uncertainty (i.e., the innovation task). Within this perspective, project teams function in the form of information-processing organisms operating in a complex and dynamic environment.

We define *effective communication* as bringing about changes in the receiver behavior that were intended by the information source (Rogers and Agarwala-Rogers 1976; Rogers and Shoemaker 1971). The intended change, in our study, relates to creating a change in knowledge or reducing uncertainty in the service innovation process, that is, reducing innovative uncertainty.

Intraproject and Extraproject Communication

We distinguish between intraproject and extraproject communication. Intraproject communication involves communication flows among members of a project team. The project team can be viewed as a task force, or "a form of horizontal contact which is designed for problems of multiple departments" (Galbraith 1974, p. 319). For instance, a task force offers a way to integrate functional viewpoints during new service development (Langeard et al. 1981). Extraproject communication involves communication flows between members of the project team and outside sources. Such flows are usually mediated by boundary spanners (e.g., Keller and Holland 1975; Tushman 1977; Tushman and Nadler 1980). Boundary spanners are individuals within the organization who serve as mediators of communication between the project team and outside sources (Allen and Cohen 1969; Keller and Holland 1975; Porter and Roberts 1983; Rothwell and Robertson 1973; Tushman 1977).

Organizational Design

To innovate successfully, the organization has to gather and process the right kind of information. Because the communication capacity of an organization is greatly influenced by its organizational structure (Bentley 1990; Gupta, Raj, and Wilemon 1986; Hage, Aiken, and Marett 1971; Souder and Moenaert 1992; Van de Ven and Delbecq 1974), structure is a critical determinant of the information processing involved in innovation activities. In this study, therefore, we pay specific attention to the organizational design in which the innovation takes place.

COMPLEXITY

Complexity is determined by the number of specialists in the organization and their professionalism (Rogers and Agarwala-Rogers 1976). The number of specialists is likely to be higher for successful organizational innovators that link greater complexity to innovation success (Gupta, Raj, and Wilemon 1986). As jobs become more complex and the variety of tasks within the organization increase, the extent and complexity of communication flows are likely to increase (Hage, Aiken, and Marrett 1971; March and Simon 1958). Therefore, innovation performance will, to an important degree, depend on the extent to which an organization can tap into diverse external knowledge sources.

As organizations grow and create more highly specialized task-related areas, these specialized subunits develop distinctive values, norms, language, and coding schemes (Katz and Kahn 1966; Thompson 1967; Tushman 1977; Tushman and Nadler 1980). When an organization has a heterogeneous background, such diversity offers an ideal platform for information exchange and can produce highly creative innovations. However, such diversity in highly complex organizations can also hinder the flow of communication that is so vital to successful innovation. Consequently, boundary spanners are needed to facilitate communication flows across subunits in complex organizations (Daft and Lengel 1986; Tushman and Nadler 1980). We expect these boundary-spanning roles to enhance extraproject communication because they will create a platform for more sources and channels of information and increase the diversity of information available within the organization. Consequently, we propose the following:

Proposition 1a: Intraproject communication relates positively to the level of complexity.
Proposition 1b: Extraproject communication relates positively to the level of complexity.

FORMALIZATION

Formalization is defined as the importance the organization attaches to the idea of following rules and procedures when performing one's job (Gupta, Raj, and Wilemon 1986). In the banking sector, both the nature of the business (i.e., cash resources, security) and the strong interdependencies that result from simultaneous production and consumption necessitate coordination and integration through formalization. In addition, in this setting and in particular in new service development, boundary-spanning activities play a role by transferring extraorganizational and intraorganizational communication and connecting the innovation team to its external environment. Conflicting research findings in the existing innovation literature (e.g., Moenaert et al. 1995) suggest that there is the possibility of a curvilinear relationship between the level of formalization and project communication. Indeed, in the literature on the role of organizational structure in the integration of marketing and R&D, formalization appears to be both a facilitator and a barrier to interfunctional integration (Hage, Aiken, and Marett 1971; Moenaert and Souder 1990a; Moenaert et al. 1995; Ruekert and Walker 1987; Souder and Moenaert 1992).

According to some researchers, formalization may prohibit the search for new sources of information, which is essential during the initiation stage of the innovation process (Zaltman, Duncan, and Holbek 1973). On the other hand, a higher degree of formalization also has been shown to facilitate the implementation stage of the innovation process (Rogers and Agarwala-Rogers 1976). Also, within a banking context, higher levels of formalization have been associated with more active product innovation and bank performance (e.g., Johne and Harborne 1985; Reidenbach and Moak 1986).

Therefore, we expect that the level of formalization will be curvilinearly related (inverted U) to the transfer of intraproject information and external information mediated by boundary spanners. The following propositions are deduced:

Proposition 2a: Intraproject communication relates curvilinearly (inverted U) to the level of formalization.

Proposition 2b: Extraproject communication (i.e., project members acting as boundary spanners) is curvilinearly related (inverted U) to the level of formalization.

CENTRALIZATION

Centralization is defined as the concentration of power in organizations and the absence of participation in decision making (Van de Ven and Ferry 1980; Walton 1981). A high degree of centralization—a high degree of hierarchy of authority combined with a low degree of participation in decision making—is likely to limit the amount of information available in the organization and restrict the channels of communication. Centralization will discourage organizational innovativeness because a high degree of hierarchy of authority limits the communication of negative feedback that might help the organization in its innovation efforts (Gupta, Raj, and Wilemon 1986; Zaltman, Duncan, and Holbek 1973). Hage, Aiken, and Marrett (1971) point to the creation of social distance through the concentration of power, thus inhibiting the rate of task communication, especially lateral communication, between departments.

In a highly centralized organization, subordinates typically must implement decisions and not participate in the development of these decisions. Centralization will therefore create a less favorable climate for project members to engage in boundary-spanning activities. Thus, we expect that a greater hierarchy of authority and a lack of participation in decision making will restrict information flows. This is postulated in the following propositions:

Proposition 3a: Intraproject communication relates negatively to the degree of centralization.

Proposition 3b: Extraproject communication relates negatively to the level of centralization.

PROJECT CLIMATE

In an innovation context, we consider the organizational climate surrounding the project, or the project climate. In a services setting, improving the organizational climate is considered to be a critical function of internal marketing (Bowen and Schneider 1988; Grönroos 1990). Thus, the quality of the project climate may strongly influence the internal communication flows. A good project climate is likely to improve the motivation of the persons involved (Mohr and Nevin 1990; Souder 1987) and to contribute to interfunctional integration and thus to innovation success (Souder and Moenaert 1992).

Findings in the R&D management literature indicate that a medium-quality project climate is best for effective project communication. As Souder (1987) has pointed out, the “too-good friends” syndrome may occur within project teams, which means that high-quality project climates may actually impede, rather than enhance, innovation success. A very high level of integration or coherence among team members can lead to too much conformity, thereby hindering the flow of a wide range of information and ideas (Janis 1973; Katz 1982; Katz and Allen 1982). Therefore, we expect that a moderate project climate is the best antecedent for enhancing the transfer of communication both within the project team and for boundary-spanning activities undertaken by project members. The following propositions are deduced:

Proposition 4a: Intraproject communication relates curvilinearly (inverted U) to the quality of the project climate.

Proposition 4b: Extraproject communication relates curvilinearly (inverted U) to the quality of the project climate.

Innovative Uncertainty Reduction: The Moderating Effect of Cross-Functional Cooperation

We have conceived the innovation process of financial services as one of uncertainty reduction. In line with Galbraith (1974), we conceive *uncertainty* as the difference between the amount of information necessary to perform a task and the amount of information already available within the organization. We expect communication to reduce the degree of uncertainty about the project, that is, to reduce the level of innovative uncertainty within the new financial service innovation team (Allen 1985; De Meyer 1985; Fidler and Johnson 1984; Moenaert and Souder 1990b; Souder and Moenaert 1992; Zaltman, Duncan, and Holbek 1973). Four types of innovative uncertainty can be distinguished: (a) uncertainty about user needs, (b) uncertainty about competitors, (c) uncertainty about technologies, and (d) uncertainty about the resources required to accomplish the financial service project. Project team members attempt to reduce the first three types of innovative uncertainty by sharing information and by building knowledge that originates from the external environment (e.g., information about customers, competitors, and technologies). The fourth type of innovative uncertainty, resource uncertainty, is related to the internal environment and deals with the question of how to allocate the appropriate financial, technical, and human resources to reduce uncertainty about customers, technologies, and competitors during the innovation task (Moenaert and Souder 1990a).

Proposition 5a: The level of innovative uncertainty reduction (i.e., about customers, competitors, technologies, and resources) relates positively to the exchange of intraproject and extraproject communication.

The need for integration and intraorganizational coordination increases in cases where the innovation process involves reciprocal task interdependencies between departments (Galbraith 1973; Thompson 1967; Tushman 1979a). Internal communication is a prime mechanism for stimulating cross-functional cooperation and for getting team members actively involved. Indeed, previous research has pointed to the positive relationship of the integration of functionally specialized activities and innovation success (e.g., Cooper 1979; Griffin and Hauser 1994; Maidique and Zirger 1984; Moenaert et al. 1995; Pinto and Pinto 1990; Van de Ven and Delbecq 1974). Thus, taking into account the importance of integration and its impact on innovation performance, we expect that the effectiveness of project communication in reducing innovative uncertainty will be contingent upon the level of cross-functional cooperation.

Moreover, we examine the boundary-spanning activities of project members who are concerned with mediating communication between their work unit and other work units within the organization. These individuals have also been called integrators (Daft and Lengel 1986; Galbraith 1974; Lawrence and Lorsch 1967) because they facilitate communication across organizational boundaries.

As a result, it is likely that the greater the level of cross-functional cooperation that has been achieved, the more effective intraproject and extraproject communication will be in reducing the level of innovative uncertainty. We can deduce the following proposition:

Proposition 5b: The level of cross-functional cooperation positively moderates the relationship between the level of innovative uncertainty reduction and the exchange of extraproject and intraproject communication.

Performance of the Financial Service Innovation Project

Innovation-related research not focusing on the R&D organization has also adopted a contingency perspective when examining the relationship between uncertainty, coordination means, and organizational effectiveness (Argote 1982; Blau and McKinley 1979; Gupta, Dirsmith, and Fogarty 1994). Based on the information-processing view that we adopted on organizational innovation, we link the level of innovative uncertainty reduction to innovation success (e.g., Allen 1985; Clark and Fujimoto 1991; Souder and Moenaert 1992). In line with this contingency-based research (e.g., Craig and Hart 1992; Griffin and Page 1993; Montoya-Weiss and Calantone 1994), we can define success as a multidimensional construct that involves financial and technological success and the achievement of learning effects. Project teams gain knowledge, or learn (e.g., Levitt and March 1988), as they exchange information. Therefore, we can define project communication effectiveness as a change in knowledge. Stated otherwise, the intended change in project members' cognitive behavior can be conceived as a reduction in the level of innovative uncertainty. We can then formulate the following proposition:

Proposition 6: Performance of the financial service innovation project relates positively to the degree of uncertainty reduction about (a) customers, (b) competitors, (c) technologies, (d) and resources.

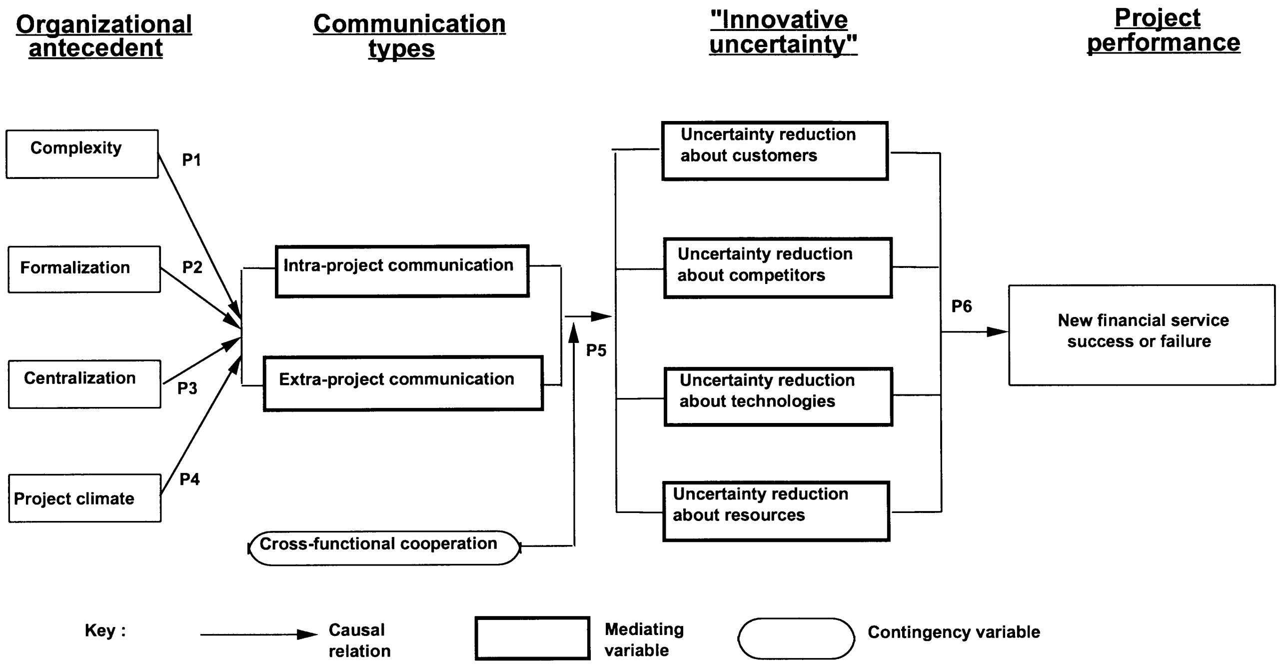
To simplify and summarize the propositions, Figure 1 describes the organizational antecedents and effects of project communication during the new financial service innovation process.

RESEARCH DESIGN

Operationalization of the Variables

The constructs of our research (see Figure 1) were operationalized according to approaches recommended by Churchill (1979), Peter (1979, 1981), and Peter and Churchill (1986). Through an intensive literature analysis and the development of a propositional framework, we specified the domain of the constructs and generated a sample of items that could be used for measurement purposes (Churchill 1979, p. 66). To create multiple-item scales for each construct, we relied on three sources of information: (a) existing operationalizations in the literature, (b) descriptions resulting from in-depth interviews designed for exploratory purposes (Lievens, Moenaert, and S'Jegers 1999), and (c) scales of our own design in cases where constructs lacked valid operationalizations in the two former sources. Two types of measurement scales were used: (a) 5-point Likert-type scales ranging from 1 = *strongly disagree* to 5 = *strongly agree* and (b) 5-point rating scales that use specific response labels (e.g., 1 = *very poor insight* to 5 = *very good insight*).

FIGURE 1
Propositional Framework on the Antecedents and Effects of Project Communication During the New Financial Services Innovation Process



Appendix A provides an overview of the constructs together with the items comprising each construct and the results of the reliability analysis. For each construct, we also briefly refer to the source or sources used for the operationalizations. All constructs show acceptable to high reliability coefficients, using an alpha level of .5 as the cutoff value for exploratory research (Cronbach and Meehl 1967; Nunnally 1967). For the centralization, complexity, and learning effects constructs, we improved and further purified the scales by deleting items that showed low interitem correlations.

Project success, that is, the performance of the financial service innovation, was assessed using several measures. This is in line with the generally held view that success is a multidimensional construct (Craig and Hart 1992; Griffin and Page 1993; Montoya-Weiss and Calantone 1994). Both financial and nonfinancial indicators were used; specifically, three performance measures have been included: financial performance, technological performance, and the achievement of learning effects (Clark and Fujimoto 1991). Financial success was included because it predominates in the new product (Craig and Hart 1992) and new service literature (e.g., de Brentani 1995a; Storey and Easingwood 1993). Because there is as yet relatively little information about the technological nature of new financial service innovations and its impact on success, we were particularly interested in technological performance in a financial service innovation context. In addition, previous case studies (Lievens, Moenaert, and S'Jegers 1999) had indicated that some innovation projects may be viewed as unsuccessful from a commercial point of view but may be seen as generating important learning effects for future innovation projects. Therefore, we decided to include this performance measure as well.

An Ex Post Facto Design: Survey Research

A survey was organized to analyze the antecedents and the impact of effective project communication on project success. Our unit of analysis is the new financial service innovation project.

THE QUESTIONNAIRE

Prior to data collection, a pretest of the survey instrument was organized in two leading Belgian banks (involving a total of 20 financial service innovation projects). The purpose of this pretest was (a) to assess construct validity and further purify the scales if necessary and (b) to evaluate and improve the quality of the questionnaire prior to full implementation of the survey. We developed a structured survey instrument involving closed-end questions.

Two key contact persons in each bank encouraged respondents to provide additional remarks and comments on the questionnaire, concerning (a) time needed to fill out the questionnaire, (b) issues related to questions and scaling, (c) difficulty and flow of the questions, and (d) wording. We also organized two in-depth interviews with two pretest respondents to assess the following concerns about internal validity:

1. We evaluated potential demand characteristics during the pretest, that is, respondents' efforts to try to anticipate research objectives (Campbell and Stanley 1963; Judd, Smith, and Kidder 1991).
2. With regard to the scale, we checked (a) potential halo bias (Cooper and de Brentani 1984; Judd, Smith, and Kidder 1991; Nunnally and Bernstein 1994, p. 373), which refers to the rater's tendency to systematically perceive an individual as being high (or low) in one area and thus being high (or low) in other areas as well, and (b) social desirability (Nunnally and Bernstein 1994), or the rater's tendency to provide ratings that reflect socially desirable behaviors.

SAMPLING STRATEGY

The sampling frame of our study was based on the Belgian Banking Association directory of 124 banks and savings institutions operating in Belgium. To reach key informants, we usually addressed the letter requesting participation to the commercial director of the bank or to the marketing manager. This initial sampling frame was revised following a first round of telephone calls through which we eliminated those banks not engaged in new service development activities. The revised sampling frame consisted of 77 banks and saving institutions.

We asked respondents in the participating financial institutions to select two completed service innovation projects. In the questionnaire, the term *project* was used to define the task or process of transforming a service idea into a new service launched in the market. To maximize the research validity, we asked respondents to choose projects based on the following criteria: (a) one project is a commercially successful service innovation, the other project is a commercially unsuccessful service innovation; (b) the two projects had been completed recently, that is, market launch occurred a maximum of 2 years before; and (c) the projects are largely internally developed projects (not fully outsourced) and involved personnel from different departments within the organization. Typically, respondents were project managers and marketing and segment managers, as well as product managers involved in the development of the new financial service innovations.

Out of a total of 154 questionnaires (two projects for all 77 banks in our revised sample frame), we achieved a response rate of 42%. Of these 65 valid questionnaires, returned by 36 banks, 37 were commercially successful projects and 28 were commercial failures. This bias on the success side has been shown to be the norm in cross-sectional surveys dealing with new financial service development (e.g., Cooper and de Brentani 1991; de Brentani 1989, 1991, 1993a, 1993b, 1995a, 1995b; Easingwood and Storey 1993, 1994; Edgett and Parkinson 1994). Several strategies and follow-up efforts were undertaken to bring the response to the level achieved. The mailing consisted of an introductory letter, a faxed document to identify the respondents and the corresponding project, and two questionnaires with accompanying letters for these respondents. Anonymity and confidentiality were ensured. Participants were told in an accompanying letter that they would benefit from involvement in the study. The participating banks received two structured questionnaires with mainly closed questions. A total of four follow-up rounds by telephone were organized, allowing a time span of 2 weeks between the follow-up sessions. A final reminder letter with questionnaires was sent to the potential respondents exactly 1 month after the last follow-up telephone round. Finally, the participating banks received a summary report and were invited to a presentation and feedback session held by the researchers.

DATA ANALYSIS

Data Screening

Because all of our hypotheses are directional, the significance tests used were one-tailed. Given the limited size of our sample, we decided to set the alpha level of significance at .1 (Type I error) to minimize Type II errors (Cook and Campbell 1979; Hays 1986; Henkel 1976). Prior to the data analysis, all variables were carefully screened to check assumptions of normality, linearity, independence, and homogeneity of variance (Tabachnick and Fidell 1989).

Through the inspection of the standardized scores (measures with a z score in excess of $\pm$ 3.16) and the normal probability plots, univariate outliers were identified among the continuous variables. Multivariate outliers were identified through computation of Mahalanobis's distance. To minimize the impact on the small sample size, we decided not to delete these cases but to reduce their in-

TABLE 1
Correlation Matrix Between Organizational Antecedent Variables and Project Communication Variables

Variable	*Mean*	SD	*Complexity*	*Formalization*	*Centralization*	*Project Climate*	*Intraproject Communication*	*Extraproject Communication*
Complexity	3.94	.71						
Formalization	3.50	.84	.42****					
Centralization	2.31	.88	–.33**	–.28*				
Project climate	3.79	.93	.30**	.19*	–.39**			
Intraproject communication	3.72	.61	**.59******	**.69******	**–.51******	**.51******		
Extraproject communication	3.46	.74	**.66******	**.40******	**–.37****	**.27****	.56****	

NOTE: Table entries represent Pearson product moment correlations. Using pairwise deletion, the sample sizes vary between 64 and 65. Hypothesized correlations are in bold print.
*$p < .1$. **$p < .05$. ****$p < .001$ (all one-tailed).

fluence by assigning them a value that was one unit larger (or one unit smaller) than the next extreme score in the distribution that was not an outlier (Tabachnick and Fidell 1989, p. 70).

A test of homogeneity of variance, using both Cochran's *C* and Bartlett's *F* tests (see Hays 1986) to check the comparability of the failed and successful projects, was performed. Results of these two tests indicated that there was no difference between the group of successful and failed projects, suggesting the appropriateness of testing for differences in mean values.

Research Findings

Testing our propositional framework enables us to shed more light on project communication as a mediator (Baron and Kenny 1986) in our causal framework concerning the nature of the link with its antecedents and its impact on project performance through the level of innovative uncertainty reduction. Project communication was labeled a mediator variable (Baron and Kenny 1986) because its role changes from a dependent variable (in its relationship to the antecedent organizational variables) to an independent variable (in terms of its contribution to uncertainty reduction). The results of the analyses are presented below. Appendix B comprises the entire correlation matrix involving all constructs under study.

ORGANIZATIONAL ANTECEDENTS OF PROJECT COMMUNICATION

Table 1 presents the bivariate correlation coefficients (Pearson zero-order correlation coefficients), the means, and the standard deviations for the constructs under consideration. Proposition 1 is supported by the findings. The level of complexity relates positively both to intraproject team communication ($r = .59$, $p < .001$) and to extraproject

TABLE 2
The Antecedent Role of Complexity and Centralization on Project Communication: Multiple Regression Analyses

	Intraproject Communication	*Extraproject Communication*
Complexity	.465***	.606***
Centralization	–.356**	–.170*
R^2 adjusted	.434	.446
F	25.11	26.40
Significance	< .0001	< .0001

NOTE: Intraproject and extraproject communication are dependent variables and represent column entries. Complexity and centralization are independent variables and represent row entries. Table entries represent standardized beta coefficients. $N = 65$. Listwise deletion of missing data.
*$p < .1$. **$p < .05$. ***$p < .01$ (all one-tailed).

communication (i.e., project members' boundary-spanning activities) ($r = .66$, $p < .001$). Proposition 3 is also supported. As hypothesized, intraproject communication ($r = –.51$, $p < .001$) and the project members' boundary-spanning activities ($r = –.37$, $p < .05$) show a negative relation with the degree of centralization.

Using the level of complexity and centralization as regressors, two standard multiple regressions were run with intraproject and extraproject as the dependent variables, respectively. The statistically significant regressions of intraproject communication ($R^2 = .43$) and extraproject communication ($R^2 = .45$) on complexity and centralization are significant and are reported in Table 2. Both complexity ($\beta = .47$, $p < .001$) and centralization ($\beta = –.36$, $p < .05$) are significant predictors of intraproject communication. Higher complexity stimulates communication within the project team, as opposed to higher levels of centralization, which hinders the communication flows within the project team. Moreover, complexity ($\beta = .61$, $p < .01$) was shown to significantly contribute, whereas cen-

TABLE 3
The Antecedent Role of Formalization and Project Climate on Project Communication: Hierarchical Curvilinear Regression Analyses

	Intraproject Communication	*Extraproject Communication*		*Intraproject Communication*	*Extraproject Communication*
Formalization	.724	.523	Project climate	.671	.344
Formalization^2	.088	–.297**	Project climate^2	–.292**	.141
R^2 adjusted	.465	.149	R^2 adjusted	.249	.056
R^2 change	.006	.074	R^2 change	.060	.014
F	55.82	11.99	F	22.265	4.724
Significance	< .0001	< .001	Significance	< .0001	< .05
F change	.759	5.886	F change	5.471	.924
Significance	*ns*	< .05	Significance	< .05	*ns*

NOTE: Four bivariate regression equations involving two regressors (formalization and project climate) linked to two regressands (intraproject communication and extraproject communication). Quadratic terms of the independent variables (regressors) were entered after the variables from which they were calculated (original variable) were entered into the regression equation. Table entries represent standardized beta coefficients. $N = 65$. Listwise deletion of missing data. One-tailed probabilities of the beta coefficients: $*p < .1$, $**p < .05$, $***p < .01$.

tralization ($\beta = –.17$, $p < .1$) significantly impeded the exchange of intraorganizational information by project members.

To test the curvilinear relationships (formulated in Propositions 2 and 4, which relate the level of formalization and project climate to intraproject and extraproject communication, respectively), second-degree polynomial (i.e., the independent variable is raised to a certain power) regression equations were used. This entailed adding to the four bivariate regression equations (the two communication variables as the regressands with the two organizational variables as regressors) the quadratic term of the independent variables in question (i.e., formalization and project climate) (see Table 3). Through hierarchical regression analysis, the proportion of variance of the dependent variable, incremented by the quadratic vector, was tested for statistical significance (Pedhazur 1982). Proposition 2 is partly supported by the data. The level of formalization is curvilinearly related (inverted U) to the transfer of extraproject information by project members. These findings indicate that moderately formalized project environments are the most favorable antecedent condition for organizational liaisons to transfer extraproject information. Further analysis of Table 1 indicates that formalization may be linearly related to intraproject communication ($r = .69$, $p < .001$), considering the significant correlation coefficient. Partial support is also found for Proposition 4. In line with our expectations, a curvilinear relationship appears to exists between the quality of the project climate and intraproject communication. Both relatively low-quality and high-quality project climates seem to create barriers for information transfer within the project team. In other words, a moderate project climate again is shown to be the best condition for intraproject communication. No support was found for a curvilinear relationship between project climate and the boundary-spanning activities of project members. Based on Table 1, the latter may be linearly related ($r = .27$, $p < .05$) to project climate.

Project Communication Effectiveness

According to the correlation coefficients presented in Table 4, Proposition 5 is largely supported ( ). Intraproject communication was found to have a positive impact on the level of uncertainty reduction about customers ($r = .48$, $p < .001$), technologies ($r = .32$, $p < .05$), resources ($r = .40$, $p < .001$), and competitors ($r = .21$, $p < .1$). Similarly, the results indicate that extraproject communication contributes to uncertainty reduction about competitors ($r = .35$, $p < .05$), customers ($r = .35$, $p < .05$), and technologies ($r = .24$, $p < .05$).

Using the level of intraproject communication and extraproject communication as independent variables in multiple regressions relating to the four types of innovative uncertainty reduction (i.e., dependent variables) (see Table 5), the R^2s were found to be statistically significant in all cases: (a) customer uncertainty ($R^2 = .20$), (b) competitive uncertainty ($R^2 = .10$), (c) resource uncertainty ($R^2 = .14$), and (d) technological uncertainty ($R^2 = .08$), specifically. Extraproject communication by project members contributed significantly only to uncertainty reduction about competitors ($\beta = .38$, $p < .05$), whereas intraproject communication contributed to the reductions in customer uncertainty ($\beta = .38$, $p < .01$), technological uncertainty ($\beta = .28$, $p < .1$), and resource uncertainty ($\beta = .46$, $p < .01$). However, in contrast to what was hypothesized in Proposition 5a and indicated by the correlation analyses, the two types of communication were not both always significantly indicated for each form of uncertainty reduction. This can partly be explained by the relatively

TABLE 4
Correlation Matrix Between the Project Communication Variables and Innovative Uncertainty Variables

Variable	*Mean*	SD	*Intraproject Communication*	*Extraproject Communication*	*Customer Uncertainty*	*Competitive Uncertainty*	*Technological Uncertainty*	*Resource Uncertainty*
Intraproject communication	3.72	.61						
Extraproject communication	3.46	.74	.56****					
Customer uncertainty	3.47	.72	**.48********	**.35****				
Competitive uncertainty	3.03	1.18	**.21***	**.35****	.24**			
Technological uncertainty	3.34	.93	**.32****	**.24****	.25**	.42****		
Resource uncertainty	3.51	.88	**.40********	.15	.19*	.37**	.64****	

NOTE: Table entries represent Pearson product moment correlations. Using pairwise deletion, the sample sizes vary between 62 and 65. Hypothesized and significant correlations are in bold print.
*p < .10. **p < .05. ****p < .001 (all one-tailed).

high intercorrelation (r = .56) of the two independent variables.

THE MODERATING ROLE OF CROSS-FUNCTIONAL COOPERATION

Proposition 5b is supported. We performed hierarchical regression analysis to examine the moderating impact of the level of cross-functional cooperation on project communication effectiveness. The significant interactions are presented in Table 6. Thus, our findings support the contingency hypothesis that we formulated: Project communication effectiveness depends on the level of cross-functional cooperation achieved. The strong interdependencies that exist due to the specific nature of services require interaction and integration (e.g., Mahajan et al. 1994; Thompson 1967). Cross-functional cooperation seems essential for enhancing this integration. Remarkable is its moderating role in reducing resource uncertainty, regardless of the type of communication concerned. Although our data did not provide support for the relationship between boundary spanners and the reduction of resource uncertainty, cross-functional cooperation seems to be a crucial moderator in enhancing this relationship. Moreover, the interaction terms of cross-functional cooperation with intraproject communication significantly enhances uncertainty reduction about competitors and technologies. The interaction of boundary-spanning communication and cross-functional cooperation also contributes to uncertainty reduction about customers.

Two additional analyses provide further validation of the pivotal but moderating role of cross-functional cooperation on project communication effectiveness. One approach involved eliminating cross-functional cooperation as a potential moderator in the relationship between innovative uncertainty reduction and project performance. Hierarchical regression analyses of the performance measures were performed (i.e., financial success, technological success, and learning effects) on the innovative uncertainty domains and the level of cross-functional cooperation.

TABLE 5
Effective Project Communication: Multiple Regression Analyses

	Customer Uncertainty	*Competitive Uncertainty*	*Technological Uncertainty*	*Resource Uncertainty*
Intraproject communication	.381***	–.036	.279*	.460***
Extraproject communication	.142	.375**	.082	–.109
R^2 adjusted	.200	.098	.080	.139
F	8.899	4.429	3.650	6.122
Significance	< .0001	< .05	< .05	< .05

SOURCE: Dependent variables represent column entries (customer uncertainty, competitive uncertainty, resource uncertainty, and technological uncertainty). Independent variables represent the row entries (intraproject communication and extraproject communication). Table entries represent standardized beta coefficients. N = 65. Listwise deletion of missing data.
*p < .1. **p < .05. ***p < .01 (all one-tailed).

The cross-functional cooperation variable was entered only after the innovative uncertainty domains had been included. In this way, we could test whether the increment in the proportion of variance in the performance measures, accounted for by the innovative uncertainty variables, was statistically significant. In line with our expectations, we found that cross-functional cooperation did not significantly increase R^2; that is, it does not appear to contribute to new financial services success. Considering that the order in which predictors are entered in the hierarchical regression strongly influences results, we performed a standard multiple regression of the performance measures on the innovative uncertainty domains and the level of cross-functional cooperation. Again, cross-functional cooperation did not emerge as a significant predictor.

TABLE 6
Statistically Significant Hierarchical Regression of the Moderating Effects of Cross-Functional Cooperation

Interaction Terms (cross-functional cooperation with . . .)	*Sign Interaction Term*	*Dependent Variable*	R^2 *Adjusted*	R^2 *Change*	F *Change*	*Significance* F *Change*
Intraproject communication	Positive	Competitive uncertainty	.012	.06	3.776	$p < .1$
Intraproject communication	Positive	Technological uncertainty	.096	.053	3.796	$p < .1$
Intraproject communication	Positive	Resource uncertainty	.155	.095	7.999	$p < .01$
Extraproject communication	Positive	Competitive uncertainty	.099	.047	3.423	$p < .1$
Extraproject communication	Positive	Resource uncertainty	.138	.104	8.223	$p < .01$

NOTE: Interaction terms (with cross-functional cooperation) were entered after the variables from which they were calculated were entered into the regression equation. $N = 65$. Listwise deletion of missing data. One-tailed probabilities.

Our data also demonstrate the mediating role of the level of innovative uncertainty reduction between cross-functional cooperation and project performance. In other words, they provide support for the proposition that the effect of cross-functional cooperation on project performance is not a direct one but one that is mediated by the level of innovative uncertainty reduction (Baron and Kenny 1986). As Table 7 illustrates, significant positive correlations exist between cross-functional cooperation and the degree of resource uncertainty reduction ($r = .38$, $p < .05$), technological uncertainty reduction ($r = .33$, $p < .05$), and customer uncertainty reduction ($r = .30$, $p < .05$). Similarly, there are significant positive correlations between cross-functional cooperation and financial performance ($r = .25$, $p < .05$), technological performance ($r = .36$, $p < .05$), and learning effects ($r = .26$, $p < .05$). We postulate that these significant correlations result mainly from the mediating role of innovative uncertainty reduction. Indeed, this assertion was supported after calculating the partial correlation between cross-functional cooperation and the performance measures, controlling for the level of innovative uncertainty reduced. In two of three cases (i.e., financial and technological success), correlations between cross-functional cooperation and project performance were no longer significant. Remarkable and interesting, the correlation with learning effects remained significant ($r = .23$, $p < .05$) after partialing the effect of innovative uncertainty reduction.

Project Success

Proposition 6 is partly supported. Table 7 gives the Pearson correlation coefficients, the mean, and the standard deviation for the involved constructs. The extent of customer uncertainty reduced ($r = .34$, $p < .05$) and the amount of resource uncertainty reduced ($r = .21$, $p < .1$) positively and significantly correlate with financial success. Technological success relates positively to the reduction of customer uncertainty ($r = .31$, $p < .05$), resource uncertainty ($r = .46$, $p < .001$), and technological uncertainty ($r = .42$, $p < .001$). The reduction of competitive uncertainty does not show a significant relationship with any of the performance measures. The feedback group discussions suggested that one plausible explanation may reside in the transparency of the market structure within the Belgian banking industry. At that time, information about competitors was widely available. This may explain why competitive uncertainty reduction seems less likely to be a priority during the innovation task. However, this situation may of course be different for that of other countries in which market structure is less stable. As a result, these data have to be interpreted with caution.

Our findings also do not show any significant correlations between the reduction of innovative uncertainty and learning effects. However, the findings suggest that the level of cross-functional cooperation between departments is an important contributor to the creation of learning effects.

MANAGEMENT IMPLICATIONS AND FUTURE RESEARCH

Innovation management within the financial industries has to realize that financial processes are complex and knowledge intensive and require much use of information technologies. Moreover, strong interdependencies exist between marketing, operations, and other departments, due to the organization-client interface. Therefore, cooperation and integration between these departments is an important antecedent of new financial service success. As de Brentani (1989) stated, “Service firms also have difficulties getting their technical and systems people (e.g., computer experts, process specialists) and their marketing personnel (e.g., frontline or services concept specialists) to effectively combine their respective points of view and proficiencies” (p. 250). Future research questions may relate to the type of departments involved in service innova-

TABLE 7
Correlation Matrix Between Innovative Uncertainty Domains, Performance Measures, and Cross-Functional Cooperation

Variable	*Mean*	SD	*Financial Performance*	*Technological Performance*	*Learning Effects*	*Customer Uncertainty*	*Competitive Uncertainty*	*Technological Uncertainty*	*Resource Uncertainty*	*Cross-Functional Cooperation*
Financial performance	2.82	1.30								
Technological performance	3.32	1.24	.65****							
Learning effects	3.55	.83	.53****	.31**						
Customer uncertainty	3.47	.72	**.34****	**.31****	.16					
Competitive uncertainty	3.03	1.18	–.13	.12	–.08	.24**				
Technological uncertainty	3.34	.93	–.01	**.42******	–.03	.25**	.42****			
Resource uncertainty	3.51	.88	**.21***	**.46******	.12	.19*	.37**	.64****		
Cross-functional cooperation	3.84	.80	.25**	.36**	.26**	.30**	.12	.33**	.38**	

NOTE: Table entries represent Pearson product moment correlations. Using pairwise deletion, the sample sizes vary between 61 and 65. Hypothesized and significant correlations are in bold print.
*$p < .1$. **$p < .05$. ****$p < .001$ (all one-tailed).

tion as well as the marketing/R&D counterpart within a services setting, and financial services setting in particular.

In fact, project managers have to deal with sometimes complex marketing interfaces during new service development. These interfaces primarily involve the management of horizontal processes within the organization. Project managers have to keep careful track of these processes, knowing the activities, flows, and interactions between the departments involved; noticing delays; and taking action whenever necessary. However, the latter involves the coordination of a complex process in which the boundaries of the involved functional departments must be crossed and connections between them have to be made in order for service innovation to be successful. As a result, the staffing of cross-functional teams is crucial for project success in general and, as our findings indicate, for learning, in particular, to take place.

In fact, the contribution to organizational learning as such should be a continuous aim of the innovation effort. It is exactly through the specialist knowledge and expertise of the professional staff that financial service organizations can build a competitive advantage. The enhanced learning through cross-functional cooperation is a type of core competence that is much harder to copy, thereby reducing the threat of competition. Therefore, innovation management should consider the importance of learning-project teams. These teams should not only benefit from previous experience, but they should also learn from other teams previously or currently engaged in similar tasks. Consequently, future research should examine cross-functional teams' contributions to learning during new service development.

Therefore, managers responsible for new service development need to recognize the critical communication roles that project members fulfill, because these individuals' intraproject communication flows foster uncertainty reduction during the innovation process and ultimately affect the level of project success. Moreover, boundary spanners link the project team to the external sources of information that are crucial for successfully fulfilling the innovation task.

A question that has not been addressed in this research is, indeed, Who are these boundary spanners? and Do they have any unique characteristics? We could assume that these integrators are eager to learn, are strong communicators, and are able to link and integrate different viewpoints between the involved departments. Depending on the specific information needs, different project team members could perform boundary-spanning activities. Frontline employees may channel important customer information to the project team. Process engineers or computer experts together with highly skilled knowledge workers could act as boundary spanners regarding essential competitive and technological information. Moreover, team structure may be adapted during the project life cycle to allow for participation by representatives of other organizations (e.g., affiliated companies, universities, etc.). Being reliable, discrete, cooperative, flexible, and independent may be some of the characteristics to investigate in the future when researching the optimal profile of boundary spanners. Again, the latter may be contingent upon the type of information required during the project life cycle.

Thus, a balanced approach is needed when managing intraproject and extraproject communication in order to preserve diversity and heterogeneity within the organization because this has been shown to enhance innovative uncertainty reduction and, ultimately, project performance. The strong correlation between intraproject and extraproject communication suggests the necessity of this balanced approach. Future research, however, could probe more deeply into the possible trade-off between boundary-spanning communication and intraproject communication during the project life cycle. As Stork (1991) stated, "Individuals' communication patterns will become increasingly constrained by the groups of which they are members" (p. 180). Due to individual information-processing limits, individuals try to deal with the problem of information overload by considering a limited set of interaction partners within the organization. Thus, individuals tend to establish strong intragroup communication, whereas boundary-spanning and intergroup communication tend to decrease (Granovetter 1973; Stork 1991).

This trade-off may be steered by innovation management's designing organizational characteristics for effective innovation. As our findings suggest, managers should create a flexible, facilitative, participative, harmonious culture in which project team members work. The climate should be supportive and encouraging to project team members. Innovative behavior—the willingness to experiment, learn, and take risks—should be rewarded and appreciated. New service teams may be reinforced in nonmonetary ways as well. Project team members may be praised for their work on the generation and implementation of innovation ideas. Also bonuses for executives may be linked to the number of new service introductions (Drew 1994).

However, managers also have to be aware of the "too-good friends" syndrome (Katz and Allen 1982). Therefore, team-based mechanisms in which the composition of the team and the nature of cooperation with others are regularly changed, supported by boundary-spanning communication, may prevent too much integration. Therefore, project management must fine-tune organizational design in order to successfully guide the information-processing activities of the project team during the innovation process. Decentralizing project structures may boil down to the stimulation and empowerment of project team members to deal with development problems themselves rather than to rely on top management. Decentralizing innovation management through the participation of operational staff (i.e., frontline personnel) may strongly influence involvement and employee satisfaction. As such, decentralizing innovation policies may eventually enhance market share and customer satisfaction through improved service delivery. A flexible project-team environment that enhances creativity and the flow of relevant information mediated by boundary spanners requires a moderately formalized organizational setting. Our findings suggest that project managers should consider a mix of formal and informal communication during service development. Although our research is quite clear on the existence of a curvilinear relationship, corroboration in future research will have to deal with the following questions: Is there a benchmark for project teams or organizations? What is the optimum in terms of organizational design? What is the threshold against which project communication becomes ineffective? Moreover, future research should address the type of project structures that may create and enhance favorable design contingencies for organizing new financial service development.

CONCLUSION

In this study, we assessed the antecedents and the effects of project communication and their impact on the performance of the financial service project. Communication was examined from an information-processing perspective, that is, in terms of uncertainty reduction. Support was found for the contribution of uncertainty reduction to the financial and technological success of new financial services. Our study also provides additional empirical evidence that the appropriate organizational design is an important prerequisite for effective project communication. The level of complexity and formalization increases intraproject communication, whereas the level of centralization appears to be a barrier for communication within the project team. A curvilinear relationship between the quality of the project climate and intraproject communication indicates that a moderate project climate is the best condition for intraproject communication. The better the quality of the project climate and the more complex the project environment, the more extraproject information will be transferred by project members. However, moderately formalized project environments seem to provide the most favorable conditions for boundary-spanning communication to occur in new service development.

All four types of innovative uncertainty were found to be significantly reduced through intraproject communication. Moreover, except for resource uncertainty, boundary-spanning communication was also an important contributor to the development of a knowledge base. Our findings reflected the crucial role of cross-functional cooperation as a contingency variable in ensuring project communication effectiveness. The greater the degree of cross-functional cooperation, the more effective intraproject and extraproject communication are in reducing the level of innovative uncertainty. When examining the information-processing impact of communication, the reduction of customer uncertainty and resource uncertainty were found to be significant predictors of both financial and technological success, whereas technological uncertainty reduction was found to contribute significantly to the perceived level of technological success.

APPENDIX A
Individual Items Used to Measure the Constructs (with reliability estimates)

Project Success: The Performance of the New Financial Service

New financial service performance has been assessed by several measures. This is in line with the generally held view that success is a multidimensional construct (Craig and Hart 1992; Griffin and Page 1993; Montoya-Weiss and Calantone 1994). Financial and nonfinancial indicators were used to measure success. These measures were selected on the basis of (a) internal consistency (i.e., if the item pools were generated on the basis of existing measures) and (b) case study research. The following scale instruction was provided on the questionnaire:

> Shown below are a number of statements and questions. Along with each statement, you will find a scale. Please indicate how strongly you agree with each of the following statements, relative to the presently discussed project. Circle the scale value that best reflects your opinion. (1 = *strongly disagree* to 5 = *strongly agree*)

Financial performance (Cronbach's $\alpha = .96$). We developed a multiple-item scale for financial performance that reflects the market share, sales growth, and overall profitability of the new financial service. The scale is based on de Brentani's (1989, 1991) research on financial service innovation.

1. This project achieved the initial commercial objectives and expectations.
2. This new service exceeded market share objectives.
3. This new service exceeded sales growth objectives.
4. Overall profitability of this new service was high.

Learning effects (Cronbach's $\alpha = .87$). Following the case study research findings, we designed our own scale for learning effects. Project teams are seen primarily as information-processing systems. In line with our definition of communication and communication effectiveness, we consider the effect of communication to be cognitive, equivalent to the creation of a knowledge base (i.e., uncertainty reduction). As a result, project teams gain knowledge and consequently learn (e.g., Kim 1993; Leonard-Barton 1995; Levitt and March, 1988) as they exchange information.

1. Our experience and learning in this project proved to be essential for the successful creation and completion of subsequent projects.
2. The knowledge acquired during the innovation process of this project served as an essential input for other new service developments.
3. The development of this new financial service created a general development expertise that eased the development and introduction of subsequent new services.
4. The experience of developing and launching this new financial service lead to enhanced know-how for future innovation projects.
5. Through the development and launch of this new service, project members learned a lot on new financial service innovation.

Technological Success

We decided to assess technological success by a single item: "How would you describe the technological success of this project?" This was rated on a 5-point scale (1 = *very unsuccessful* to 5 = *very successful*). This selection was based on in-depth discussions with pretest respondents. They considered this item to better represent the construct of technological success. Although this might contradict the view within measurement theory that multiple-item scales are more reliable (Churchill 1979), Peter and Churchill (1986, p. 4) stated that measure characteristics affect construct validity not only through reliability but also via content validity.

Innovative Uncertainty Domains

The measures concerning the innovative uncertainty domains were based on the operationalization of Moenaert and Souder (1990a) and were partly adapted for financial service innovation projects (Lievens, Moenaert, and S'Jegers 1999). The following scale instruction was provided on the questionnaire:

> Shown below are some items concerning the type of information available during development and launch of this project. Indicate for each item how well you and other project members were informed during the innovation process of this new service. The choice runs from 1 (*no knowledge existed for that subject*) to 5 (*we knew everything we had to know*). Values between 1 and 5 reflect different levels of knowledge in the respective areas. Please circle the number that best reflects your choice.

Competitive uncertainty (Cronbach's α = .85).

1. The technological strategy of the competition.
2. The marketing strategy of the competition.

Customer uncertainty (Cronbach's α = .74).

1. The customer's needs (user requirements).
2. The potential market.
3. The buyer behavior of the potential customer.

Technological uncertainty (Cronbach's α = .91).

1. The quality of the applied technologies (e.g., information technologies).
2. The user-friendliness of the technologies.
3. The cost-efficiency of the technologies.

Resource uncertainty (Cronbach's α = .80).

1. The required R&D strategy for this project.
2. The required technological support for this project.
3. The required personnel for this project.

Project Communication

To specify the nature of the information that is transferred during the financial service innovation process, we used the communication typology proposed by Hauptman (1986): coordinative and innovative information. Hauptman examined this communication typology as a mediator between task type and performance in the case of software development. The duality between the coordination and innovation requirements of the task was proposed as a predictor of optimal communication patterns and was linked to the structure, stability, and newness of the technology involved in the task. Both types of information give a clear picture of the kind of information that is exchanged during the innovation process. Innovative communication comprises creativity in problem solving and new idea processing and adaptation to change. Coordinative communication relates to the assignment of instructions to project members so that they can execute their jobs (Souder and Moenaert 1992) and includes controls, orders, direction, and feedback between subordinates and superiors in task-related activities (Greenbaum 1974). Boundary-spanning communication by project members was based on a scale of our own design, partly derived from the literature (e.g., Tushman 1977).

Intraproject communication (Cronbach's α = .70). The following scale instruction was provided on the questionnaire:

> Shown below are a number of statements concerning communication during the innovation process. Under each statement, you will find a numbered scale and an explanation of the appropriate meaning of the numbers. Please rate each statement according to the degree the statement describes your opinion of this project. Draw a circle around the number that best reflects your choice.

1. During the development and launch preparation of this project, how good was project team members' insight of what everybody else involved with this project was actually doing? (1 = *very poor insight* to 5 = *very good insight*)
2. To what extent do you agree with the following statement: "Information concerning the technological requirements was widely available during new service development." (1 = *strongly disagree* to 5 = *strongly agree*)
3. During the innovation process of this project, how well were project members informed and updated concern-

ing commercial information (customer, competition)? (1 = *very poorly* to 5 = *excellent*)

4. Less structured information is also exchanged during new service development in order to induce the innovativeness of the project in which new information is generated or problems are solved in a creative way. How frequently was information transferred that was helpful in solving work-related problems? (1 = *basically never* to 5 = *once a day*)

Extraproject communication (Cronbach's α .83). The following scale instruction was provided on the questionnaire:

Shown below are a number of statements and questions. Along with each statement, you will find a scale. Please indicate how strongly you agree with each of the following statements, relative to the presently discussed project. Circle the scale value that best reflects your opinion. (1 = *strongly disagree* to 5 = *strongly agree*)

1. The development of this project was characterized by interdepartmental communication stimulated by project members who had regular contacts with their colleagues from other departments.
2. The project team members involved with this new service were particularly active in communicating with personnel outside their departments.
3. Some team members of this project acted as liaisons between different departments by transferring information and thereby fostering cooperation between them.
4. Through the efforts of some project members, this innovation process was characterized by a steady exchange of information between the departments.

Organizational Variables

The following scale instruction was provided on the questionnaire:

Shown below are a number of statements concerning the organizational structure in which the new service was developed. Along with each statement, you will find a scale. Please indicate how strongly you agree with each of the following statements, relative to the presently discussed project. Circle the scale value that best reflects your opinion. (1 = *strongly disagree* to 5 = *strongly agree*)

Cross functional cooperation (Cronbach's α = .90). Cross-functional cooperation was adapted from Mahajan et al. (1994) and Pinto, Pinto, and Prescott (1993).

1. This project showed active involvement of all parties involved.
2. Cooperation between project members was characterized by a steady and fluent exchange of resources (money, personnel, equipment, office space).
3. Work flow (materials, objects, and customers) was easily transferred between the project members involved.
4. An excellent interaction was present between the members involved for the transfer of information.

Complexity (Cronbach's α = .80). Complexity was partly based on the operationalization of Gupta and Wilemon (1986).

1. This project involved cooperation from different functional specialists.
2. The development of this new service was characterized by the involvement of a wide range of specialist skills including marketing, operations, and technical personnel.
3. The project required skills and professional inputs from a diversity of functional departments.
4. The project members involved during development of this project were selected because of their specialist skills within their respective fields.

Formalization (Cronbach's α = .82). We assessed formalization through existing measures (Dewar, Whetten, and Boje 1980; Gupta, Raj, and Wilemon 1986; Ruekert and Walker 1987; Walton 1981).

1. The development and launch preparation was totally unstructured: everybody was allowed to do almost as he or she pleased. (R)
2. There were precise time schedules for the activities to be undertaken for this project.
3. Project team members were given very specific job instructions.
4. This project proceeded by means of a well-documented action plan.

Centralization (Cronbach's α = .55). Centralization was assessed through existing measures (Dewar, Whetten, and Boje 1980; Gupta, Raj, and Wilemon 1986; Robert and O'Reilly 1974; Walton 1981).

1. Top management delegated responsibilities for the tasks involving this project. (R)
2. Top management encouraged project personnel to make suggestions. (R)

Project climate (Cronbach's α = .92). The operationalization of project climate was based on previous operationalizations (Mohr and Nevin 1990; Souder and Moenaert 1992) as well as on previous case study findings (Lievens, Moenaert and S'Jegers 1999). For example, in developing a measure for project climate, we used the cited formulations given by project managers during the in-depth interviews: "a positive attitude, a supportive climate, a high level of enthusiasm, a strong commitment of personnel and management."

1. The whole new service development project took place in a harmonious climate.
2. The relationship among the project members during service development and launch preparations was very good.
3. The project members involved in this project trusted one another.
4. The relationship between the project members was excellent.

NOTE: (R) = items were reverse coded.

APPENDIX B
Entire Correlation Matrix of the Constructs Under Study

Variable	*Complexity*	*Formalization*	*Centralization*	*Project Climate*	*Intraproject Communication*	*Extraproject Communication*	*Customer Uncertainty*	*Competitive Uncertainty*	*Technological Uncertainty*	*Resource Uncertainty*	*Financial Performance*	*Technological Performance*	*Learning Effects*
Complexity													
Formalization	.42****												
Centralization	–.33**	–.28*											
Project climate	.30**	.19*	–.39**										
Intraproject communication	.59****	.69****	–.51****	.51****									
Extraproject communication	.66****	.40****	–.37****	.27**	.56***								
Customer uncertainty	.32**	.34**	.13	.09	.48****	.35**							
Competitive uncertainty	.37**	.08	.15	.02	.21*	.35**	.24**						
Technological uncertainty	.40**	.26*	.24*	.35**	.32**	.24**	.25**	.42****					
Resource uncertainty	.28*	.29**	.36**	.35**	.40****	.15	.19*	.37**	.64****				
Financial performance	.26*	.27*	.17	.14	.37**	.25*	.34**	–.13	–.01	.21*			
Technological performance	.33**	.25*	.31**	.31**	.49**	.26*	.31**	.12	.42****	.46****	.65****		
Learning effects	.32**	.21*	.34**	.20	.38**	.23*	.16	–.08	–.03	.12	.53****	.31**	
Cross-functional cooperation	.45**	.39**	.56**	.75**	.69**	.37**	.30**	.12	.33****	.38**	.25**	.36**	.26**

NOTE: Table entries represent Pearson product moment correlations. Using pairwise deletion, the sample sizes vary between 61 and 65.
*$p < .1$. **$p < .05$. ****$p < .001$ (all one-tailed).

REFERENCES

Allen, T. J. (1985), *Managing the Flow of Technology: Technology Transfer and the Dissemination of Technological Information within the R&D Organization*. Cambridge, UK: MIT Press.

——— and S. I. Cohen (1969), "Information Flow in Research and Development Laboratories," *Administrative Science Quarterly*, 14 (1), 12-19.

Argote, L. (1982), "Input Uncertainty and Organizational Coordination in Hospital Emergency Units," *Administrative Science Quarterly*, 27 (September), 420-34.

Baron, R. M. and D. A. Kenny (1986), "The Moderator-Mediator Variable Distinction in Social Psychological Research: Conceptual, Strategic, and Statistical Considerations," *Journal of Personality and Social Psychology*, 51 (6), 1173-82.

Bentley, K. (1990), "A Discussion of the Link between One Organization's Style and Structure and Its Connection with Its Market," *Journal of Product Innovation Management*, 7, 19-34.

Blau, J. R. and W. McKinley (1979), "Ideas, Complexity, and Innovation," *Administrative Science Quarterly*, 24 (June), 200-18.

Bowen, D. E. and B. Schneider (1988), "Services Marketing and Management: Implications for Organizational Behavior," *Research in Organizational Behavior*, 10, 43-80.

Brown, J. W. and J. M. Utterback (1985), "Uncertainty and Technical Communication Patterns," *Management Science*, 31 (3), 301-11.

Campbell, D. T. and J. C. Stanley (1963), *Experimental and Quasi-experimental Designs for Research*. Dallas: Houghton Mifflin.

Chase, R. B. (1981), "The Customer Contact Approach to Services: Theoretical Bases and Practical Extensions," *Operations Research*, 29, 698-706.

——— and D. A. Tansik (1983), "The Customer Contact Model for Organization Design," *Management Science*, 29 (9), 1037-50.

Churchill, G. A. Jr. (1979), "A Paradigm for Developing Better Measures of Marketing Constructs," *Journal of Marketing Research*, 16 (February), 64-73.

Clark, K. B. and T. Fujimoto (1991), *Product Development Performance: Strategy, Organization and Management in the World Auto Industry*. Boston: Harvard Business School Press.

Cook, T. D. and D. T. Campbell (1979), *Quasi-Experimentantation: Design and Analysis Issues for Foeld Settings*. Boston: Houghton Mifflin.

Cooper, R. G. (1979), "The Dimensions of Industrial New Product Success and Failure," *Journal of Marketing*, 43 (Summer), 93-103.

——— and U. de Brentani (1984), "Criteria for Screening New Industrial Products," *Industrial Marketing Management*, 13, 149-56.

——— and ——— (1991), "New Industrial Financial Services: What Distinguishes the Winners," *Journal of Product Innovation Management*, 8, 75-90.

———, C. J. Easingwood, S. Edgett, E. J. Kleinschmidt, and C. Storey (1994), "What Distinguishes the Top Performing New Products in Financial Services," *Journal of Product Innovation Management*, 11, 281-99.

Craig, A. and S. Hart (1992), "Where to Now in New Product Development Research?" *European Journal of Marketing*, 26 (11), 3-36.

Cronbach, L. J. and P. E. Meehl (1967), "Construct Validity in Psychological Tests," in *Problems in Human Assessment*, D. N. Jackson and S. Messic, eds. New York: McGraw-Hill, 57-77.

Daft, R. L. and R. H. Lengel (1986), "Organizational Information Requirements, Media Richness and Structural Design," *Management Science*, 32 (5), 554-71.

——— and K. E. Weick (1984), "Towards a Model of Organizations as Interpretation Systems," *Academy of Management Review*, 9 (2), 284-95.

de Brentani, U. (1989), "Success and Failure in New Industrial Services," *Journal of Product Innovation Management*, 6, 239-58.

——— (1991), "Success Factors in Developing New Business Services," *European Journal of Marketing*, 24 (2), 33-59.

——— (1993a), "The Effect of Customization and Customer Contact on the Factors Determining Success and Failure of New Industrial Services," in *Perspectives on Marketing Management*, Vol. 3, M. J. Baker, ed. New York: John Wiley, 275-98.

——— (1993b), "The New Product Process in Financial Services: Strategy for Success," *International Journal of Bank Marketing*, 11 (3), 15-22.

——— (1995a), "Firm Size: Implications for Achieving Success in New Industrial Services," *Journal of Marketing Management*, 11, 207-25.

——— (1995b), "New Industrial Service Development: Scenarios for Success and Failure," *Journal of Business Research*, 32, 93-103.

——— and R. G. Cooper (1991), "New Industrial Financial Services: What Distinguishes the Winners?" *Journal of Product Innovation Management*, 8, 75-90.

——— and ——— (1992), "Developing Successful New Financial Services for Businesses," *Industrial Marketing Management*, 21, 231-41.

De Meyer, A.C.L. (1985), "The Flow of Technological Innovation in an R&D Department," *Research Policy*, 14, 315-28.

Dewar, R. D., D. A. Whetten, and D. Boje (1980), "An Examination of the Reliability and Validity of the Aiken and Hage Scales of Centralization, Formalization, and Task Routineness," *Administrative Science Quarterly*, 25 (March), 120-28.

Drew, S. (1994), *Business Re-Engineering in Financial Services*. London: Pitman.

Easingwood, C. J. (1986), "New Product Development For Service Companies," *Journal of Product Innovation Management*, 4, 264-75.

——— and J. Percival (1990), "Evaluation of New Financial Services. International," *Journal of Bank Marketing*, 8 (6), 3-8.

——— and C. Storey (1991), "Success Factors for New Consumer Financial Services," *International Journal of Bank Marketing*, 1, 3-10.

——— and C. Storey (1993), "Marketplace Success Factors for New Financial Services," *Journal of Services Marketing*, 7 (1), 41-54.

——— and C. Storey (1994), "The Value of Multi-Channel Distribution Strategy," in *Marketing: Its Dynamics and Challenges*, 23rd EMAC Conference, J. Bloemer, J. Lemmink, and H. Kasper, eds. Maastricht, the Netherlands: University of Limburg, 203-12.

Ebadi, Y. M. and J. M. Utterback (1984), "The Effects of Communication on Technological Innovation," *Management Science*, 30 (5), 572-85.

Edgett, S. and S. Parkinson (1994), "The Development of New Financial Services: Identifying Determinants of Success and Failure," *International Journal of Service Industry Management*, 5 (4), 24-38.

Fidler, L. A. and J. D. Johnson (1984), "Communication and Innovation Implementation," *Academy of Management Review*, 9 (4), 704-11.

Galbraith, J. R. (1973), *Designing Complex Organizations*. Reading, MA: Addison-Wesley.

——— (1974), "Organization Design: An Information Processing View," in *Organizational Psychology*, 2nd ed., D. A. Kolb, I. M. Rubin, and J. M. McIntyre, eds. Englewood Cliffs, NJ: Prentice-Hall, 313-22.

Granovetter, M. S. (1973), "The Strength of Weak Ties," *American Journal of Sociology*, 78 (6), 1361-80.

Greenbaum, H. H. (1974), "The Audit of Organizational Communication," *Academy of Management Journal*, 17 (4), 739-54.

Griffin, A. and J. R. Hauser (1994), "Integrating Mechanisms for Marketing and R&D," ISBM report, Pennsylvania State University.

——— and A. L. Page (1993), "An Interim Report on Measuring Product Development Success and Failure," *Journal of Product Innovation Management*, 10, 291-308.

Grönroos, C. (1990), *Service Management and Marketing: Managing the Moments of Truth in Service Competition*. Lexington, MA: Lexington Books.

Gupta, A. K., S. P. Raj, and D. Wilemon (1986), "A Model for Studying R&D-Marketing Interface in the Product Innovation Process," *Journal of Marketing*, 50 (April), 7-17.

Gupta, P. P., M. W. Dirsmith, and T. J. Fogarty (1994), "Coordination and Control in a Government Agency: Contingency and Institutional

Theory Perspectives on GAO Audits," *Administrative Science Quarterly*, 39, 264-84.

Hage, J., M. Aiken and C. B. Marett (1971), "Organization Structure and Communications," *American Sociological Review*, 36 (October), 860-71.

Hauptman, O. (1986), "Influence of Task Type on the Relationship between Communication and Performance: The Case of Software Development," *R&D Management*, 16 (2), 127-39.

Hays, W. L. (1986), *Statistics*, 4th ed. New York: Holt, Rinehart and Winston.

Henkel, R. E. (1976), *Tests of Significance*. Beverly Hills, CA: Sage.

Janis, I. L. (1973), *Victims of Groupthink: A Psychological Study of Foreign Policy Decisions and Fiascoes*. Boston: Houghton Mifflin.

Johne, F. A. and P. Harborne (1985), "How Large Commercial Banks Manage Product Innovation," *International Journal of Bank Marketing*, 3 (1), 54-70.

Judd, C. M., E. R. Smith, and L. H. Kidder (1991), *Research Methods in Social Relations*, 6th ed. Forth Worth, TX: Harcourt Brace Jovanovich.

Katz, D. and R. L. Kahn (1966), *The Social Psychology of Organizations*. New York: John Wiley.

Katz, R. (1982), "The Effects of Group Longevity on Project Communication and Performance," *Administrative Science Quarterly*, 27 (1), 81-104.

——— and T. J. Allen (1982), "Investigating the Not Invented Here Syndrome: A Look at the Performance, Tenure, and Communication Patterns of 50 R&D Project Groups," *R&D Management*, 12 (1), 7-19.

Keller, R. T. and W. E. Holland (1975), "Boundary-Spanning Activity and Research and Development Management: A Comparative Study," *IEEE Transactions on Engineering Management*, EM-22(4) (November), 130-33.

Kim, D. H. (1993), "The Link between Individual and Organizational Learning," *Sloan Management Review*, Fall, 37-50.

Langeard, E., J.E.G. Bateson, C. H. Lovelock, and P. Eiglier (1981), "Integrating Different Functional Perspectives in Service Businesses," in *Services Marketing: New Insights from Consumers and Managers*, E. Langeard, J.E.G. Bateson, C. H. Lovelock, and P. Eiglier, eds. Cambridge, MA: Marketing Science Institute, 81-95.

Lawrence, P. and J. Lorsch (1967), *Organizations and Environment*. Cambridge, MA: Harvard University Press.

Leonard-Barton, D. (1995), *Wellsprings of Knowledge: Building and Sustaining the Sources of Innovation*. Boston: Harvard Business School Press.

Levitt, B. and J. G. March (1988), "Organizational Learning," *Annual Review of Sociology*, 14, 319-40.

Lievens, A., K. de Ruyter, and J. Lemmink (1999), "Learning During New Banking Service Development: A Communication Network Approach to Marketing Departments," *Journal of Service Research*, 2 (2), 145-63.

———, R. K. Moenaert, and R. S'Jegers (1999), "Linking Communication to Innovation Success in the Financial Services Industry: A Case Study Analysis," *International Journal of Service Industry Management*, 10 (1), 23-47.

Lovelock, C. H., E. Langeard, J.E.G. Bateson, and P. Eiglier (1988), "Some Organizational Problems Facing Marketing in the Service Sector," in *Managing Services: Marketing, Operations and Human Resources*, C. H. Lovelock, ed. London: Prentice-Hall, 359-66.

Macdonald, S. (1998), *Information for Innovation: Managing Change from an Information Perspective*. Oxford, UK: Oxford University Press.

Mahajan, J., A. J. Vakharia, P. Paul, and R. B. Chase (1994), "An Exploratory Investigation of the Interdependence between Marketing and Operations Functions in Service Firms," *International Journal of Research in Marketing*, 11, 1-15.

Maidique, M. A. and B. J. Zirger (1984), "A Study of Success and Failure in Product Innovation: The Case of the U.S. Electronics Industry," *IEEE Transactions on Engineering Management*, EM31(4) (November), 192-203.

March, J. G. and H. A. Simon (1958), *Organizations*. New York: John Wiley.

Moenaert, R. K. and W. E. Souder (1990a), "An Analysis of the Use of Extrafunctional Information by R&D and Marketing: Review and Model," *Journal of Product Innovation Management*, 7, 213-39.

——— and ——— (1990b), "An Information Transfer Model for Integrating Marketing and R&D Personnel in New Product Development Projects," *Journal of Product Innovation Management*, 7, 91-107.

———, A. De Meyer, W. E. Souder, and D. Deschoolmeester (1995), "R&R/Marketing Communication during the Fuzzy Front-End," *IEEE Transactions on Engineering Management*, 42 (3), 243-58.

Mohr, J. and J. R. Nevin (1990), "Communication Strategies in Marketing Channels: A Theoretical Perspective," *Journal of Marketing*, (October), 36-51.

Montoya-Weiss, M. M. and R. Calantone (1994), "Determinants of New Product Performance: A Review and Meta-Analysis," *Journal of Product Innovation Management*, 11, 397-417.

Nunnally, J. C. (1967), *Psychometric Theory*, New York: McGraw-Hill.

——— and I. H. Bernstein (1994), *Psychometric Theory*, 3rd ed. New York: McGraw-Hill.

Pedhazur, E. J. (1982), *Multiple Regression in Behavioral Research*. Fort Worth, TX: Holt, Rinehart and Winston.

Peter, J. P. (1979), "Reliability: A Review of Psychometric Basics and Recent Marketing Practices," *Journal of Marketing Research*, 16 (February), 6-17.

——— (1981), "Construct Validity: A Review of Basic Issues and Marketing Practices," *Journal of Marketing Research*, 18 (May), 133-45.

——— and G. A. Churchill Jr. (1986), "Relationships among Research Design Choices and Psychometric Properties of Rating Scales: A Meta-Analysis," *Journal of Marketing Research*, 23 (February), 1-10.

Pinto, M. B. and J. K. Pinto (1990), "Project Team Communication and Cross-Functional Cooperation in New Program Development," *Journal of Product Innovation Management*, 7, 200-212.

———, ———, and J. E. Prescott (1993), "Antecedents and Consequences of Project Team Cross-functional Cooperation," *Management Science*, 39 (10), 1281-97.

Porter, L. W. and K. H. Roberts (1983), "Communication in Organizations," in *Handbook of Industrial and Organizational Psychology*, M. D. Dunnette, ed. New York: John Wiley, 1553-89.

Reidenbach, R. E. and D. L. Moak (1986), "Exploring Retail Bank Performance and New Product Development: A Profile of Industry Practice," *Journal of Product Innovation Management*, 3, 187-94.

Robert, K. H. and C. A. O'Reilly (1974), "Measuring Organizational Communication," *Journal of Applied Psychology*, 59 (3), 321-26.

Rogers, E. M. and R. Agarwala-Rogers (1976), *Communication in Organizations*. New York: Free Press; London: Collier Macmillan.

——— and F. F. Shoemaker (1971), *Communication of Innovations: A Cross-Cultural Approach*, 2nd ed. New York: Free Press.

Rothwell, R. and A. B. Robertson (1973), "The Role of Communication in Technological Innovation," *Research Policy*, 2, 204-25.

Ruekert, R. W. and O. C. Walker (1987), "Marketing's Interaction with Other Functional Units: A Conceptual Framework and Empirical Evidence," *Journal of Marketing*, 51 (January), 1-19.

Smith, K. G., K. A. Smith, J. D. Olian Jr., H. P. Sims, D. P. O'Bannon, and J. A. Scully (1994), "Top Management Team Demography and Process: The Role of Social Integration and Communication," *Administrative Science Quarterly*, 39 (September), 412-38.

Souder, W. E. (1987), *Managing New Product Innovations*. Lexington MA: Lexington.

——— and R. K. Moenaert (1992), "Integrating Marketing and R&D Project Personnel within Innovation Projects: An Information Uncertainty Model," *Journal of Management Studies*, 29 (4), 485-512.

Storey, C. and C. Easingwood (1993), "The Impact of the New Product Development Project on the Success of Financial Services," *Service Industries Journal*, 13 (3), 40-54.

Stork, D. (1991), "A Longitudinal Study of Communication Networks: Emergence and Evolution in a New Research Organization," *Journal of Engineering and Technology Management*, 7, 177-96.

Tabachnick, B. G. and L. S. Fidell (1989), *Using Multivariate Statistics*, 2nd ed. New York: HarperCollins.

Thompson, J. D. (1967), *Organizations in Action*. New York: McGraw-Hill.

Tushman, M. L. (1977), "Special Boundary Roles in the Innovation Process," *Administrative Science Quarterly*, 22 (December), 587-605.

——— (1979a), "Impacts of Perceived Environmental Variability on Patterns of Work Related Communication," *Academy of Management Journal*, 22 (3), 482-500.

——— (1979b), "Managing Communication Networks in R&D Laboratories," *Sloan Management Review*, Winter, 37-49.

——— (1979c), "Work Characteristics and Subunit Communication Structure: A Contingency Analysis," *Administrative Science Quarterly*, 24, 82-97.

——— and D. A. Nadler (1980), "Communication and Technical Roles in R&D Laboratories: An Information Processing Approach," *Management Sciences*, 15, 91-112.

Van de Ven, A. H. and A. L. Delbecq (1974), "A Task Contingent Model Of Work-Unit Structure," *Administrative Science Quarterly*, 19, 183-97.

——— and D. L. Ferry (1980), *Measuring and Assessing Organizations*. New York: John Wiley.

Walton, E. J. (1981), "The Comparison of Measures of Organization Structure," *Academy of Management Review*, 6 (1), 155-60.

Weick, K. E. (1969), *The Social Psychology of Organizing*. Boston: Addison-Wesley.

Zaltman, G., R. Duncan, and J. Holbek (1973), *Innovations and Organizations*. New York: John Wiley.

Annouk Lievens is a senior research fellow and assistant professor of marketing in the Department of Marketing and Marketing Research at Maastricht University. Her research interests and publications focus on financial services marketing, communication networks within financial services organizations, service innovation management, and marketing interfaces during new service development.

Rudy K. Moenaert is a professor of marketing at the University of Ghent in Belgium and Nyenrode University in the Netherlands. He has published in journals such as the *Journal of Product Innovation Management*, *R&D Management*, *Management Science*, *IEEE Transactions on Engineering Management*, and the *Journal of Management Studies*. His current research interests focus on the international management of product innovation, the management of innovation programs, the marketing of product innovations, and Internet marketing.